

Title

Protocol for a phase 2 randomized trial of a novel monoclonal antibody for malaria prophylaxis in young children in East Africa

Abstract

Background: Passive immunization with monoclonal antibodies targeting malaria parasite antigens offers a novel strategy complementing artemisinin-based prevention. Previous testing in older children under seasonal transmission showed promise; efficacy and safety in infants and younger children under intense perennial transmission remain unknown.

Methods: This phase 2 trial comprises two stages. Stage 1 involves sequential age-de-escalation and dose-escalation cohorts (children aged 5-10 years, then 5-59 months, then 5-71 months) receiving single escalating doses of monoclonal antibody L9LS (5, 10, 20, 30, or 40 mg/kg) or placebo (3:1 randomization) to assess safety and tolerability. Stage 2 randomizes 324 children aged 5-59 months to two-dose L9LS (baseline and month 6), single-dose L9LS (baseline only), or placebo, with 12-month surveillance for malaria infection via monthly blood slides. Primary efficacy endpoint: *Plasmodium falciparum* infection incidence. Primary safety endpoint: adverse event rates. Secondary endpoints include parasitaemia levels, clinical malaria episodes, and antibody levels.